

KILLARA HIGH SCHOOL
TERM 1 EXAMINATION 2002
2 UNIT MATHEMATICS

Time Allowed: 3 hours

Instructions:

Attempt all questions

All questions are of equal value

Approved calculators may be used

Start your answer to each question on a new page

Question 1 (14 marks) Sequence + Series

- (a) Which term of 16, 14, 12, 10, is -28? (2 marks)
- (b) If the 10th term of an AP is 34 and the 20th term is 84 find a and d . Hence find the 30th term. (4 marks)
- (c) Evaluate $\sum_{n=1}^{40} (10 - 3n)$ (2 marks)
- (d) How many terms of $1 + 3 + 9 + 27 + \dots$ are needed for the sum to exceed 3000? (3 marks)
- (e) If $x, 2, y$ are the first 3 terms of a GP and $x, y, 7$ are the first 3 terms of an AP, find x and y . (3 marks)

Question 2 (14 marks) Circular Trigonometry

- (a) Express in circular measure in terms of π (2 marks)
- (i) 45° (ii) 240°
- (b) Convert to degrees and minutes (2 marks)

(i) $\frac{\pi^c}{3}$

(ii) 1.93^c

(c) Find the area of the minor segment of a circle when $r = 12\text{cm}$, and $\theta = 60^\circ$ (2 marks)

(d) Without using graph paper and using $0 \leq x \leq 2\pi$ as domain, sketch the following functions on the one set of axes

(i) $y = \cos 3x$

(ii) $y = \cos \frac{x}{3}$

(4 marks)

(e) The longer "hand" of a clock is 1.06m in length. How far does its tip move in 45 minutes? (2 marks)

(f) A sprinkler is set to water a sector of a circle with an angle of 75° at the centre. If the range of the sprinkler is 5m calculate the area watered. (2 marks)

Question 3 (14 marks) Calculus

(a) Expand and simplify

(i) $\cos(x + \frac{\pi}{4})$

(2 marks)

(ii) $\tan(x - \frac{5\pi}{6})$

(2 marks)

(b) What is the period of $y = 3\sin 4x$? (1 mark)

(c) What is the amplitude of $y = 5\cos 2x$? (1 mark)

(d) Differentiate with respect to x

(i) $\sin(6 - x)$ (1 mark)

(ii) $3x - \cos 3x$ (2 marks)

(e) Integrate with respect to x

(i) $\cos(5 + x)$ (1 mark)

(ii) $x^3 - \sin(x + 2)$ (2 marks)

- (f) Find the primitive function of $3 \cos x + 2 \sin 2x$ (2 marks)

Question 4 (14 marks) Calculus and Trigonometry

- (a) Give the primitive functions of

(i) $5x^2$ (ii) $\frac{5}{x^2}$ (2 marks)

- (b) Evaluate

(i) $\int_{-2}^1 x^3 dx$ (ii) $\int_1^3 (x^2 - 2x + 3) dx$ (4 marks)

- (c) Find the first derivative of

(i) $(x + \frac{1}{x})^3$ (2 marks)

(ii) $\frac{\cos x}{3 - \sin x}$ (2 marks)

(iii) $(5x + 3)\tan x$ (2 marks)

(iv) $\frac{\tan(x^2 - 5)}{(x^2 - 5)}$ (2 marks)

Question 5 (14 marks) Integration

- (a) Find the area enclosed between the y axis the curve $x = y^2 + 1$ and the lines $y = 1$ and $y = 3$ (3 marks)

- (b) Find the volume of the solid created when the curve $y = x^2 + x$ is rotated about the x axis between $x = 1$ and $x = 2$. (3 marks)

- (c) Apply the Trapezoidal rule using three function values to calculate $\int_1^3 \frac{2}{x} dx$ (4 marks)

- (d) Find the area enclosed between $y = x^2 - 9$ and $y = x - 3$ (4 marks)

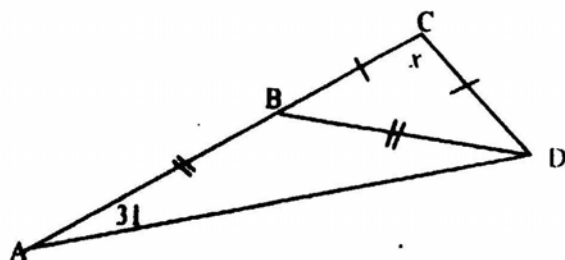
Question 6 (14 marks) Quadratic + Parabola

- (a) What values of m will make the expression $x^2 + 6x + m$ positive definite (2 marks)
- (b) Write $3x^2 + 2x + 5$ in the form $A(x - 1)^2 + B(x - 1) + C = 0$ (2 marks)
- (c) If α and β are the roots of the equation $2x^2 - 3x + 4 = 0$ find
- (i) $\alpha + \beta$
 - (ii) $\alpha\beta$
 - (iii) $\alpha^2 + \beta^2$
 - (iv) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ (4 marks)
- (d) Given $2 \tan^2 \theta - \tan \theta - 6 = 0$
- (i) Substitute $x = \tan \theta$
 - (ii) Factorise and solve for x
 - (iii) Solve for θ if $0 \leq \theta \leq 2\pi$ (4 marks)
- (e) Find the equation of the parabola with vertex $(-3, 4)$ and directrix with equation $y = 1$. (2 marks)

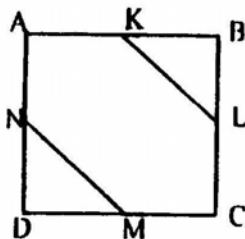
Question 7 (14 marks) Identities and Geometry

- (a) Simplify
- (i) $\frac{3}{\sec^2 \theta} + \frac{3}{\operatorname{cosec}^2 \theta}$ (1 mark)
 - (ii) $-\frac{\sec \theta}{\cos \theta} + \frac{\tan \theta}{\cot \theta}$ (1 mark)
- (b) A and B are two gun emplacements on the coastline 2 km apart. C is a target at sea such that $\angle BAC = 60^\circ 37'$ and $\angle ABC = 64^\circ 18'$. What is the range from A to target at C at sea. (4 marks)

- (c) Find the value of the pronumeral giving reasons for your answer. (3 marks)



(d)



A gift box with a square face (ABCD) is tied with ribbon crossing at K, L, M, N the midpoints of each respective side. Prove that $KL = NM$. (5 marks)

Question 8 (14 marks) Applications Derivative

- (a) For the function $f(x) = 2x^3 - 8x$ find
- the first derivative (1 mark)
 - the second derivative (1 mark)
 - hence find any stationary point(s) and explain their nature(s). (2 marks)
- (b) Find the equation of the normal, at $x = \frac{\pi}{3}$, to the curve $y = \tan(3x)$ (3 marks)
- (c) If the gradient of the tangent at the point (x, y) on a given curve is $y' = 2\sin 3x$ and the curve passes through the point $(\frac{\pi}{3}, \frac{8}{3})$ find the equation of the curve. (2 marks)
- (d) A new high school is to be built in a suburb where one side faces a road and the other 3 sides create a rectangular block. If the fencing available for these 3 sides is 700m find the length of road frontage that will give a maximum area. (3 marks)
- (e) Find the area enclosed by the curve $y = 3 \cos(x + 2)$ the x axis and the lines $x = \frac{\pi}{3}$ and $x = \frac{3\pi}{4}$ (2 marks)

Question 9 (14 marks)

- (a) Write $1.\dot{2}\dot{3}$ as a mixed fraction (2 marks)
- (b) Anna borrows \$30000 from her Credit Union @ 1.6% per month reducible interest. She repays the loan over 3 years in equal monthly instalments. What are her repayments each month? (3 marks)
- (c) Find the perpendicular distance from the point (3, 4) to the line $5x - 3y + 2 = 0$ (2 marks)
- (d) On a number plan sketch the lines
- (i) $2x + y = 4$ and (ii) $y = 3x + 6$
- (iii) Clearly label the region in which $2x + y < 4$ and $y \geq 3x + 6$ (4 marks)
- (e) Express $x^2 - 2x = 8y - 25$ in standard form and hence find
- (i) the co-ordinates of the vertex
- (ii) the co-ordinates of the focus (3 marks)

END OF PAPER

TERM 1 EXAM 2002.

2U MATHS SOLUTIONS.

QUESTION 1.

(a) $a=16$ $d=-2$. $T_n = a + (n-1)d$

$$-28 = 16 + (n-1)(-2)$$

$$-44 = -2(n-1)$$

$$n-1 = 22$$

$$n = 23$$

$\therefore -28$ is 23rd term. 1 mark

(b) $T_{10} = a + 9d = 34$

$T_{20} = a + 19d = 84$

$$\therefore 10d = 50$$

$$\therefore d = 5$$

$$\therefore a + 45 = 34$$

$$a = -11$$

OR

$$T_{20} - T_{10} = 50$$

$$\therefore T_{30} = 84$$

$$+ 50 = 134$$

$$= 134$$

$$\therefore T_{30} = a + 29d$$

$$= -11 + 29 \times 5$$

$$= 134$$

(c) $\sum_{n=1}^{40} (10 - 3n) = 7 + 4 + 1 + \dots + -110$ i.e. $a=7$ $d=-3$ 1 mark

$$S_{40} = \frac{n}{2}[2a + (n-1)d] \text{ OR } \frac{n}{2}[a+l]$$

$$= \frac{40}{2}[14 + 39(-3)]$$

$$= 20 \times -103$$

$$= -2060$$

$$= -2060$$

(d) $1+3+9+27+\dots$ $a=1$ $r=3$ i.e. GP 1 mark

$$S_n = \frac{a(r^n - 1)}{r - 1} = \frac{1(3^n - 1)}{3 - 1} > 3000$$

$$\therefore 3^n - 1 > 6000$$

$$3^n > 6001$$

$$3^8 = 6561$$

$$8 \text{ terms}$$

(e) $x, 2, y = GP$

$\therefore \frac{2}{x} = \frac{y}{2}$

$\therefore xy = 4$ — (i) | n.b.

Sub (i) into (i)

$(2y-7)y = 4$

$2y^2 - 7y - 4 = 0$

$(2y+1)(y-4) = 0$

$y = -\frac{1}{2}$ or $y = 4$

$\therefore x = -8$ or $x = 1$

ANS $(-8, -\frac{1}{2})$ or $(1, 4)$

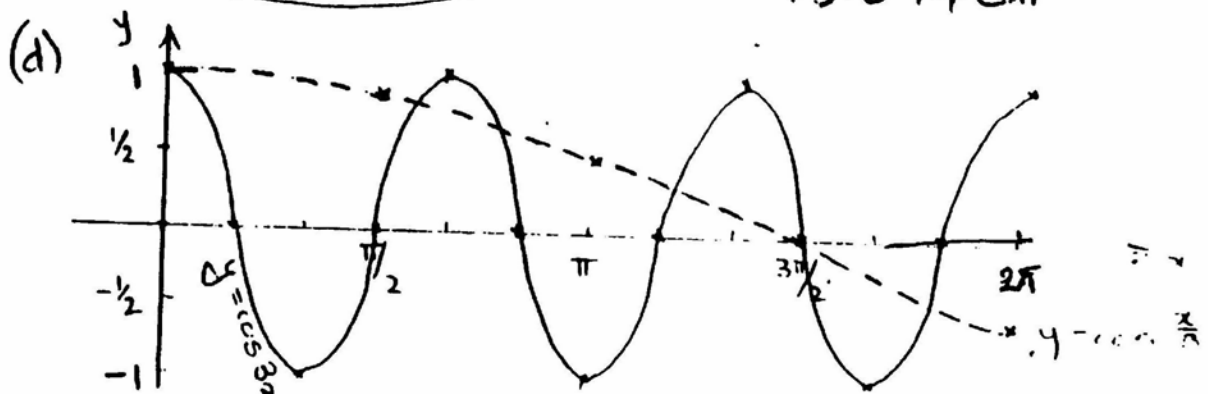
QUESTION 2.

(a) (i) $45^\circ = \frac{\pi}{4}$ | n.b. (ii) $240^\circ = \frac{4\pi}{3}$

(b) (i) $\frac{\pi}{3} = 60^\circ$ | n.b. (ii) $1.93^\circ = 110^\circ 35'$

(c) If in degrees
 $A = \frac{1}{2} \times 12^2 (60 \cdot \sin 60)$
 But = 4257.6 Allow 1

$A = \frac{1}{2} r^2 (\theta - \sin \theta)$
 $= \frac{1}{2} \times 12^2 (\frac{\pi}{3} - \sin \frac{\pi}{3})$
 $= 13.044 \text{ cm}^2$



(e) $45 \text{ mins} = \frac{3}{4} \text{ hr} \therefore \frac{3}{4} \text{ revolution or } \frac{3\pi}{2} \text{ radians}$
 $l = r\theta = 1.06 \times \frac{3\pi}{2}$
 $= 4.9951 \text{ m}$
 $\approx 5 \text{ m. } 500 \text{ cm } 4.995 \text{ mm}$

QUESTION 2 CONTINUED

$$\left. \begin{aligned} \dot{p}) \quad A &= \pi r^2 \frac{\theta}{360} \quad \text{or} \quad \frac{1}{2} r^2 \theta \\ &= \pi \times 5^2 \times \frac{75}{360} \quad \text{or} \quad \frac{1}{2} \times r^2 \times \frac{75\pi}{180} \\ &= 16.362 \text{ m}^2 \quad \text{or} \quad \frac{125\pi}{24} \text{ m}^2 \end{aligned} \right\} \quad \therefore \frac{5\pi}{12}$$

QUESTION 3.

a) ~~Ans.~~ (i) $\cos\left(\pi + \frac{\pi}{4}\right) = \cos \pi \cos \frac{\pi}{4} - \sin \pi \sin \frac{\pi}{4} \quad \text{--- Id}$
 $= \frac{\cos \pi - \sin \pi}{\sqrt{2}}$

(ii) $\tan(x - \frac{5\pi}{6})$ or $\frac{1(\cos x - \sin x)}{2}$ (Link)

$= \frac{\tan x - \tan(\frac{5\pi}{6})}{1 + \tan x \tan \frac{5\pi}{6}}$ (Link)

$= \frac{\tan x + \frac{1}{\sqrt{3}}}{1 + \tan x \times -\frac{1}{\sqrt{3}}}$ (Link)

$= \frac{\sqrt{3} \tan x + 1}{\sqrt{3} - \tan x}$ (Link)

$= \frac{\sqrt{3} \tan^2 x + 4 \tan x + 1}{3 - \tan^2 x}$ (Link)

(b) Period = $\frac{2\pi}{4} = \frac{\pi}{2}$, or 90° ... 1 mark

(C) Amplitude = 5 ——— link

(d) (i) $f(x) = \sin(6-x)$
 $f'(x) = -\cos(6-x) \quad \text{--- 1mk}$

(ii) $f(x) = 3x - \cos 3x$
 $f'(x) = 3 + 3\sin 3x \quad \text{--- 1mk}$

(e) (i) $\int \cos(5+x) dx$
 $= \sin(5+x) + C$

(ii) $\int (x^3 - \sin(x+2)) dx$
 $= \frac{1}{4}x^4 + \cos(x+2) + C$

(f) $\int (3 \cos x + 2 \sin 2x) dx = 3 \sin x - \cos 2x + C$
Omit $+C$, but once only

QUESTION 4

(a) (i) $\int 5x^2 dx = \frac{5}{3} x^3 + C$ | |

(ii) $\int 5x^{-2} dx = -5x^{-1} + C$ or $-\frac{5}{x} + C$ | |

(b) (i) $\int_{-2}^1 x^3 dx = \left[\frac{x^4}{4} \right]_{-2}^1$
 $= \frac{1}{4} - \frac{16}{4}$
 $= -3\frac{3}{4}$

(ii) $\int_1^3 (x^2 - 2x + 3) dx = \left[\frac{1}{3} x^3 - x^2 + 3x \right]_1^3$
 $= (9 - 9 + 9) - (\frac{1}{3} - 1 + 3)$
 $= 6\frac{2}{3}$

(c) (i) $\frac{d}{dx} \left(x + \frac{1}{x} \right)^3 = 3 \left(x + \frac{1}{x} \right)^2 \times \left(1 - x^{-2} \right)$
 $= 3 \left(1 - \frac{1}{x} \right) \left(x + \frac{1}{x} \right)^2$

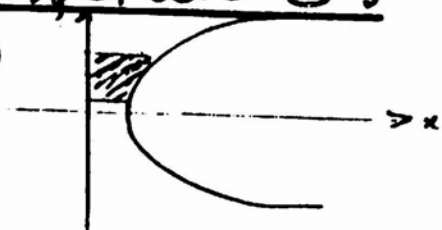
(ii) $\frac{d}{dx} \left(\frac{\cos x}{3 - \sin x} \right)$ Let $u = \cos x$ $v = 3 - \sin x$
 $u' = -\sin x$ $v' = -\cos x$
 $\therefore \frac{dy}{dx} = \frac{v u' - u v'}{v^2} = \frac{(3 - \sin x)(-\sin x) + \cos x \cos x}{(3 - \sin x)^2}$
 $= \frac{-3\sin x + \sin^2 x + \cos^2 x}{(3 - \sin x)^2}$
 $= \frac{1 - 3\sin x}{(3 - \sin x)^2}$

(iii) $y = (5x + 3) \tan x$ $u = 5x + 3$ $v = \tan x$
 $u' = 5$ $v' = \sec^2 x$
 $\therefore y' = uv' + vu' = (5x + 3) \sec^2 x + \tan x \times 5$
or $5 \tan x + (5x + 3) \sec^2 x$

(iv) $y = \frac{\tan(x^2 - 5)}{x^2 - 5}$ $u = \tan(x^2 - 5)$ $v = x^2 - 5$
 $u' = (2x) \cdot \sec^2(x^2 - 5)$ $v' = 2x$
 $y' = \frac{v u' - u v'}{v^2} = \frac{(x^2 - 5) 2x \sec^2(x^2 - 5) - 2x \tan(x^2 - 5)}{(x^2 - 5)^2}$
 $= \frac{2x [(x^2 - 5) \sec^2(x^2 - 5) - \tan(x^2 - 5)]}{(x^2 - 5)^2}$

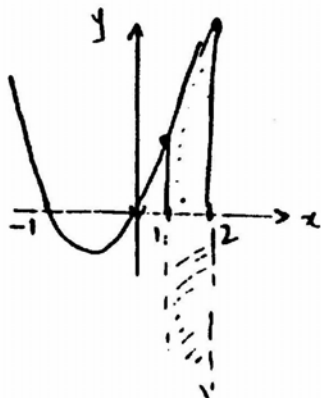
QUESTION 5:

a)



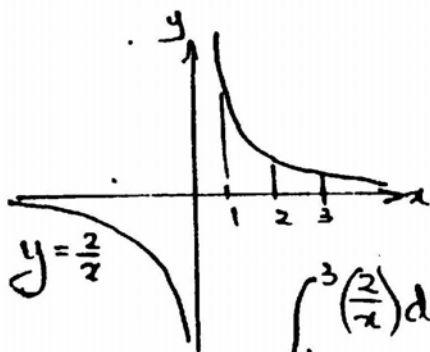
$$\begin{aligned}
 A &= \int_{-1}^3 (x^2 + 1) dy \\
 &= \int_{-1}^3 (y^2 + 1) dy \quad | \text{ or diagram} \\
 &= \left[\frac{1}{3} y^3 + y \right]_{-1}^3 \\
 &= (9 + 3) - \left(\frac{1}{3} + 1 \right) \\
 &= 10 \frac{2}{3} \text{ sq units}
 \end{aligned}$$

b)



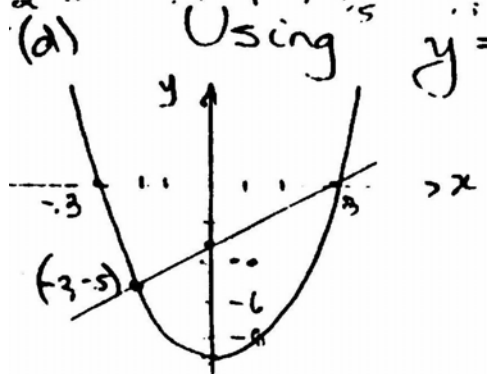
$$\begin{aligned}
 y &= x^2 + x = x(x+1) \\
 \text{Vol} &= \int_{-1}^2 \pi y^2 dx \\
 &= \pi \int_{-1}^2 (x^2 + x)^2 dx \\
 &= \pi \int_{-1}^2 (x^4 + 2x^3 + x^2) dx \\
 &= \pi \left[\frac{1}{5} x^5 + \frac{1}{2} x^4 + \frac{1}{3} x^3 \right]_{-1}^2 \\
 &= \pi \left[\left(\frac{32}{5} + 8 + \frac{8}{3} \right) - \left(-\frac{1}{5} - \frac{1}{2} - \frac{1}{3} \right) \right] \\
 &= \frac{481\pi}{30} \text{ CUB UNITS} \\
 &\div 50.37 \text{ U}^3
 \end{aligned}$$

c)



$$\begin{aligned}
 \int_1^3 \left(\frac{2}{x} \right) dx &\div \frac{b-a}{2} [f(a) + f(b)] \text{ twice} \\
 &= \frac{2-1}{2} [f(1) + f(3)] + \frac{3-2}{2} [f(2) + f(3)] \\
 &= \frac{1}{2} (2+1) + \frac{1}{2} (1+\frac{2}{3}) \\
 &= 2 \frac{1}{3} \text{ sq units}
 \end{aligned}$$

(d)



$$\begin{aligned}
 x^2 - 9 &= x - 3 \\
 \therefore x^2 - x - 6 &= 0 \\
 (x-3)(x+2) &= 0
 \end{aligned}$$

$$\begin{aligned}
 A &= | \text{Area}(x^2 - 9) - \text{Area}(x - 3) | \\
 &= \left| \int_{-3}^3 (x^2 - 9) dx \right| - \left| \int_{-3}^3 (x - 3) dx \right| \\
 &= \left| \left[\frac{1}{3} x^3 - 9x \right]_{-3}^3 \right| - \left| \left[\frac{1}{2} x^2 - 3x \right]_{-3}^3 \right| \\
 &= \left| (9 - 27) - (-\frac{8}{3} + 18) \right| - \left| \left(\frac{9}{2} - 9 \right) - \left(\frac{4}{2} + 6 \right) \right| \\
 &= | -18 + 10 \frac{2}{3} | - | -\frac{9}{2} - 6.5 | \\
 &= | -1.666... | - | -11.25 | \\
 &= 9.583...
 \end{aligned}$$

QUESTION 6.

✓ (a) $x^2 + 6x + m$ Pos D.f. If $a > 0$ $\Delta < 0$ |

$a = 1 > 0$ $\Delta = 36 - 4 \times 1 \times m < 0$

If $36 - 4m$
is $m > 9$ |

✓ (b)

$$3x^2 + 2x + 5 = A(x-1)^2 + B(x-1) + C$$

$$\text{RHS} = A(x^2 - 2x + 1) + B(x-1) + C$$

$$= Ax^2 + x(-2A+B) + (A-B+C)$$

$\therefore A = 3$ |

also $A - B + C = 5$

also $-2A + B = 2$ |

$\therefore 3 - B + C = 5$

$\therefore -6 + B = 2$

$+C = 10$ |

$\therefore B = 8$ |

Sub!

$$\therefore 3x^2 + 2x + 5 = 3(x-1)^2 + 8(x-1) + 10$$

✓ (c)

$$2x^2 - 3x + 4 = 0$$

$\therefore (i) \alpha\beta = -\frac{b}{a}$
 $= \frac{3}{2}$ |

(ii) $\alpha + \beta = \frac{c}{a}$
 $= 2$ |

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$
 $= \left(\frac{3}{2}\right)^2 - 2 \times 2$
 $= -1\frac{3}{4}$ |

(iv) $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta}$
 $= \frac{-1\frac{3}{4}}{\frac{3}{2}}$
 $= -\frac{7}{8}$ |

✓ (d)

$$2\tan^2\theta - \tan\theta - 6 = 0$$

$\therefore (i) 2x^2 - x - 6 = 0$ |

(ii) $(2x + 3)(x - 2) = 0$ |

$\therefore x = -\frac{3}{2}$ or $x = 2$ |

(iii) $\therefore \tan\theta = -\frac{3}{2}$ or $\tan\theta = 2$ |

$\theta = 56^\circ 19'$ or $303^\circ 41'$

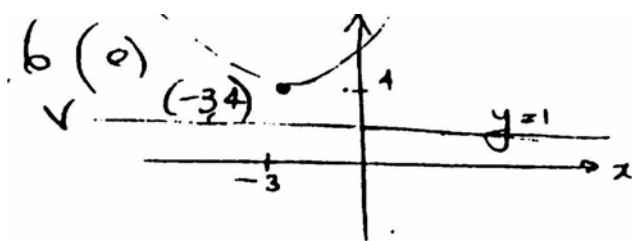
$= 123^\circ 41'$ or $303^\circ 41'$

$= 2.158$ or 5.300

$\theta = 63^\circ 26'$ or $243^\circ 26'$

$= 63^\circ 26'$ or $243^\circ 26'$

$= 1.107$ or 4.248



Focal length = 3

Parabola is of form

$$4ay = x^2$$

$$4a(y - k) = (x - h)^2$$

$$12(y - 4) = (x + 3)^2 \quad \text{or shift by 1}$$

$$12y - 48 = x^2 + 6x + 9$$

$$x^2 + 6x - 12y + 57 = 0$$

Q

QUESTION 7.

i) (i) $\frac{3}{\sec^2 \theta} + \frac{3}{\cos^2 \theta}$

$$= 3\cos^2 \theta + 3\sin^2 \theta$$

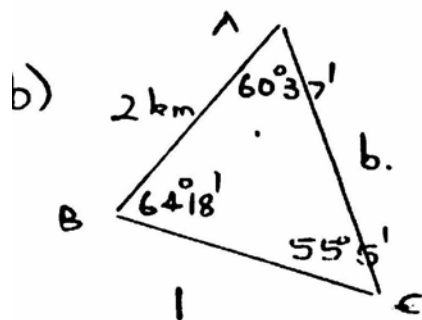
$$= 3. \quad (1)$$

(ii) $\frac{\sec \theta}{\cos \theta} + \frac{\tan \theta}{\cot \theta}$

$$= \sec^2 \theta + \tan^2 \theta$$

$$= (1 + \tan^2 \theta) + \tan^2 \theta$$

$$= 2\tan^2 \theta + 1 \quad (1)$$



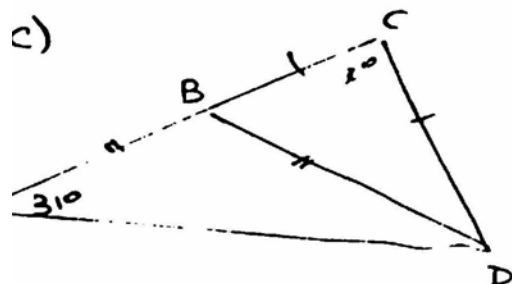
$$\frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\frac{b}{\sin 64^\circ 18'} = \frac{2000}{\sin 55^\circ 5'}$$

$$b = \frac{2000 \times \sin 64^\circ 18'}{\sin 55^\circ 5'}$$

$$= 2197.7879...$$

The target is 2198m from C
any sensible answer



-1 any omission

$$\angle BDC = 31^\circ \dots \text{L's opp} = \text{sides}$$

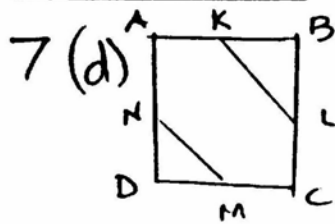
$$\angle CBD = 62^\circ \dots \text{ext } \angle \Delta = \text{sum of 2 int opp L's}$$

$$\angle CDB = 62^\circ \dots \text{L's opp} = \text{sides}$$

$$\therefore \angle BCD = 180^\circ - 124^\circ \dots \text{L sum } \Delta$$

$$x = 56$$

QUESTION 7 (CONT)



7 (d)

$AB = BC = CD = AD \dots$ ABCD is a square
 $\therefore KB = BL = DM = DN \dots$ each $\frac{1}{2}$ of above

In $\triangle DNM$ and $\triangle BKL$

$$DN = KB \dots \text{above}$$

$$DM = LB \dots \text{above}$$

$$\angle NDM = 90^\circ = \angle KBL \dots \text{ABCD is a square}$$

$$\therefore \triangle DNM \equiv \triangle BKL \dots \text{SAS}$$

$$\therefore NM = KL \dots \text{corresponding sides of congruent } \triangle's$$

QUESTION 8.

(a) $f(x) = 2x^3 - 8x$ (i) $f'(x) = 6x^2 - 8$
 (ii) $f''(x) = 12x$
 (iii) Let $f'(x) = 0 \therefore 6x^2 = 8$
 $x^2 = \frac{8}{6}$

$$x = \pm \sqrt{\frac{8}{6}}$$

$\therefore$ Stat Points are $(\frac{2}{\sqrt{3}}, f(\frac{2}{\sqrt{3}}))$

or $(\frac{2}{\sqrt{3}}, -\frac{32}{3\sqrt{3}})$

$\therefore$ Minimum

$$f''(\frac{2}{\sqrt{3}}) = \frac{24}{\sqrt{3}} > 0$$

$\therefore$ Minimum

$$(-\frac{2}{\sqrt{3}}, f(-\frac{2}{\sqrt{3}}))$$

$$(-\frac{2}{\sqrt{3}}, \frac{32}{3\sqrt{3}})$$

$\therefore$ Minimum

$$f''(-\frac{2}{\sqrt{3}}) = -\frac{24}{\sqrt{3}} < 0$$

$\therefore$ Maximum

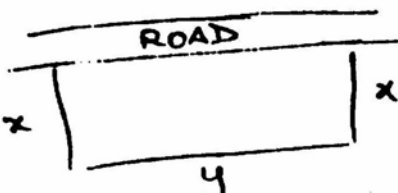
OR
 Using $f''(x)$

(b) $y = \tan 3x$
 $y' = 3 \sec^2 3x$
 $f'(\frac{\pi}{3}) = 3 \sec^2 \pi$
 $= 3(-1)^2$
 $= 3$
 $\therefore -\frac{1}{m} = -\frac{1}{3}$
 $f(\frac{\pi}{3}) = \tan \pi$
 $= 0$

Normal $y - y_1 = m(x - x_1)$
 $y - 0 = \frac{1}{3}(x - \frac{\pi}{3})$
 $y = -\frac{x}{3} + \frac{\pi}{9}$
 or $\frac{x}{3} + y = \frac{\pi}{9}$
 or $x + 3y = \frac{\pi}{3}$

(c) $y' = 2\sin 3x$
 $\therefore y = -\frac{2}{3}\cos(3x) + C.1$
 $\therefore \frac{y}{3} = -\frac{2}{3}\cos(\pi) + C.$
 $\therefore \frac{y}{3} = \frac{2}{3} + C$
 $\therefore C = 2.$

Curve $y = -\frac{2}{3}\cos(3x) + 2.$
 $= 2 - \frac{2}{3}\cos(3x)$

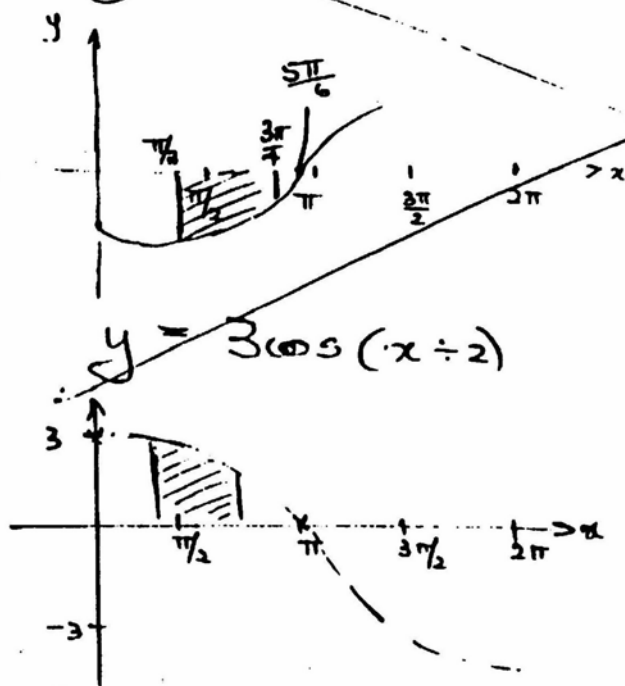
(d) 
 $2x + y = 700$
 $y = 700 - 2x.1$

$\therefore$ Road frontage
 $= 700 - 2 \times 175$
 $= 350m$

$A = x \times y$
 $= x(700 - 2x)$
 $= 700x - 2x^2$
 $A' = 700 - 4x$
For stat part
 $700 = 4x$
 $x = 175.$

$A'' = -4 < 0$
 $\therefore$ Max.

(e) $y = 3\cos(x+2)$



$A = - \int_{\pi/3}^{3\pi/4} 3\cos(x+2) dx.$
 $= -3 \int_{\pi/3}^{3\pi/4} \cos(x+2) dx$
 $= -3 \left[\sin(x+2) \right]_{\pi/3}^{3\pi/4}$
 $= -3 \left[\sin\left(\frac{3\pi}{4}+2\right) - \sin\left(\frac{\pi}{3}+2\right) \right]$
 $= -3 \left[(-0.937) - (0.09425) \right]$

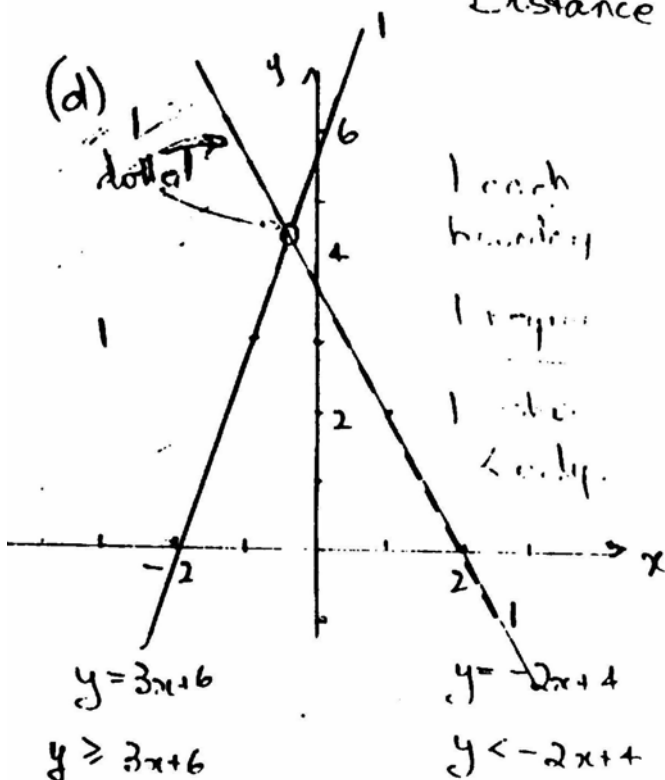
$A = \int_{\pi/3}^{3\pi/4} 3\cos\left(\frac{x}{2}\right) dx$
 $= 3 \left[2\sin\left(\frac{x}{2}\right) \right]_{\pi/3}^{3\pi/4}$
 $= 6 \left[0.9238... - 0.5 \right]$
 $= 2.542577 \div 2 = 1.2712885$

QUESTION 9

(a) $1.\dot{2}\dot{3}$ Let $x = 1.\dot{2}\dot{3}$
 $= 1\frac{23}{99} - 1$ $\therefore 100x = 123.\dot{2}\dot{3}$
 $\therefore 99x = 122$
 $x = \frac{122}{99} = 1\frac{23}{99} - 1$

(b) Borrow $A = P(1 + \frac{R}{100})^n$
 $= 30000(1.016)^{36}$
 $= 53124.48597...$
 Repays $P_1 = Q(1.016)^{35}$
 $P_2 = Q(1.016)^{34}$
 $\vdots$
 $P_{36} = Q$
 Total $S_n = Q(\frac{r^n - 1}{r - 1})$
 $= Q \frac{(1.016^{36} - 1)}{1.016 - 1}$
 $= Q \times 48.176012...$
 $\therefore Q = \frac{53124.48597...}{48.176012...} = 1102.71555...1$
 Repayment $\div$ \$1102.70 or \$1102.72 or \$1103, given

(c) $d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}} = \frac{|5 \times 3 - 3 \times 4 + 2|}{\sqrt{5^2 + 3^2}}$
 Distance $= \frac{5}{\sqrt{34}}$ or $\frac{5\sqrt{34}}{34}$



(e) $x^2 - 2x = 8y - 25$
 $x^2 - 2x + 1 = 8y - 24$
 $(x - 1)^2 = 8(y - 3)$
 (i) $a = 2$ Vertex $(1, 3)$

(ii) $\therefore$ Focus $(1, 5)$